



U.S. General Services
Administration

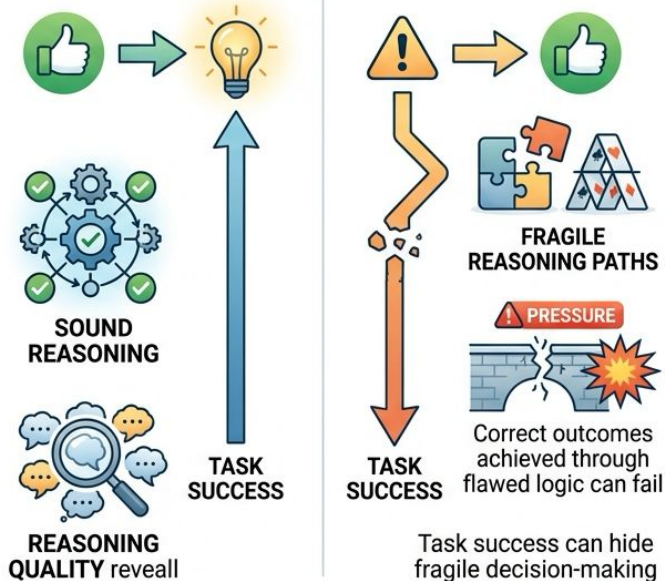
Chapter 3.9: Reasoning Quality

US General Service Administration
AI Community of Practice - March 2026

Five Dimensions of Reasoning Quality

Agent Performance: Sound vs. Fragile Reasoning

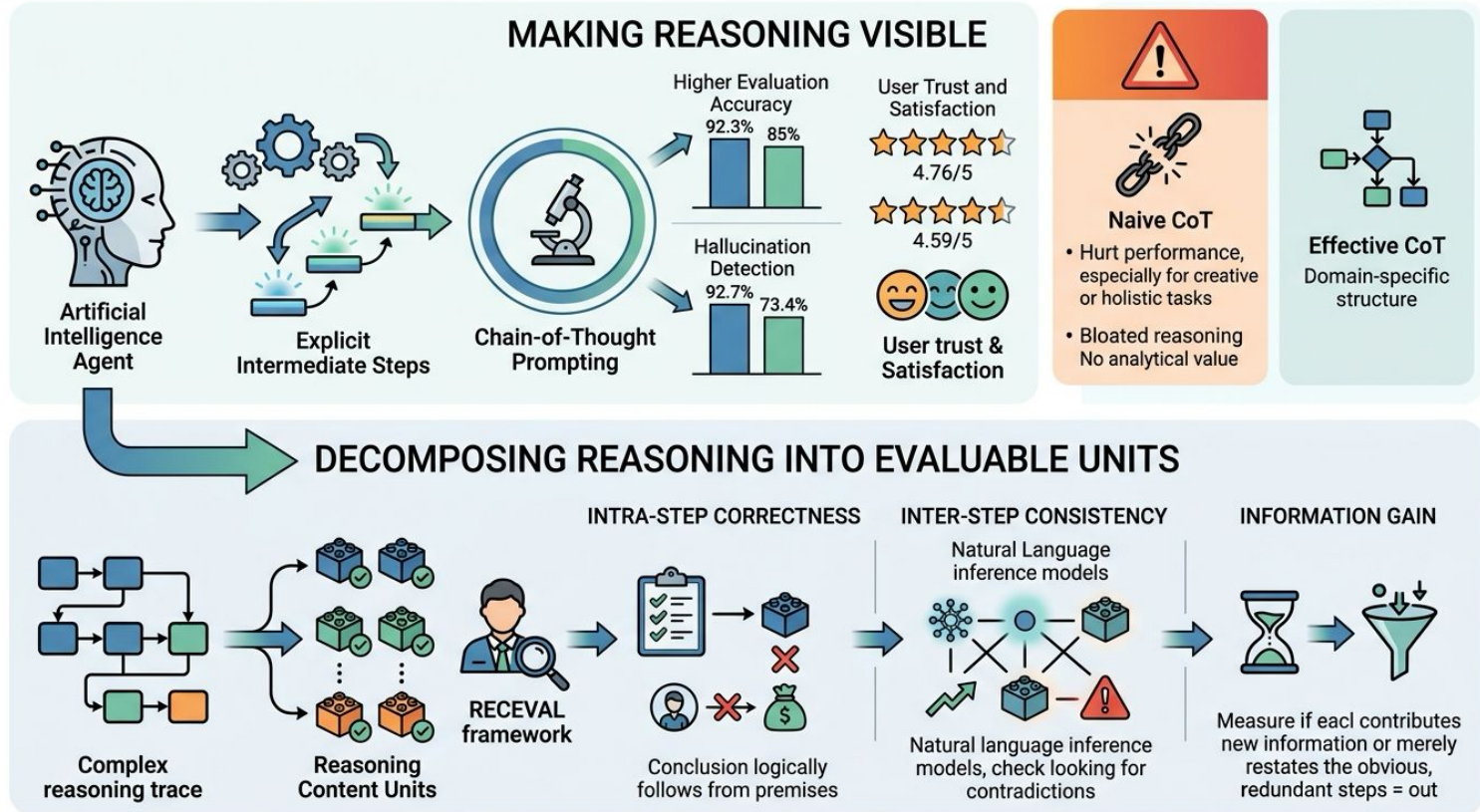
Right Answers, Wrong Reasons



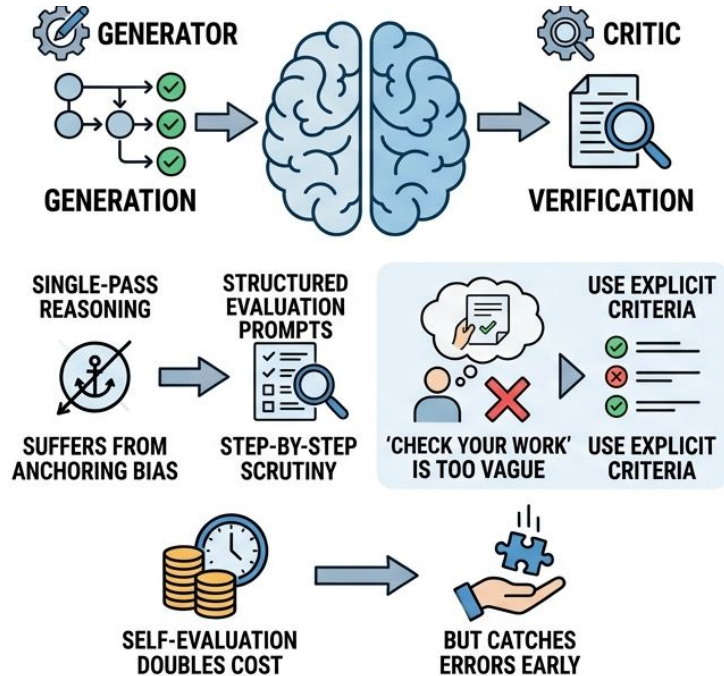
Five Dimensions of Reasoning Quality



Making Reasoning Visible



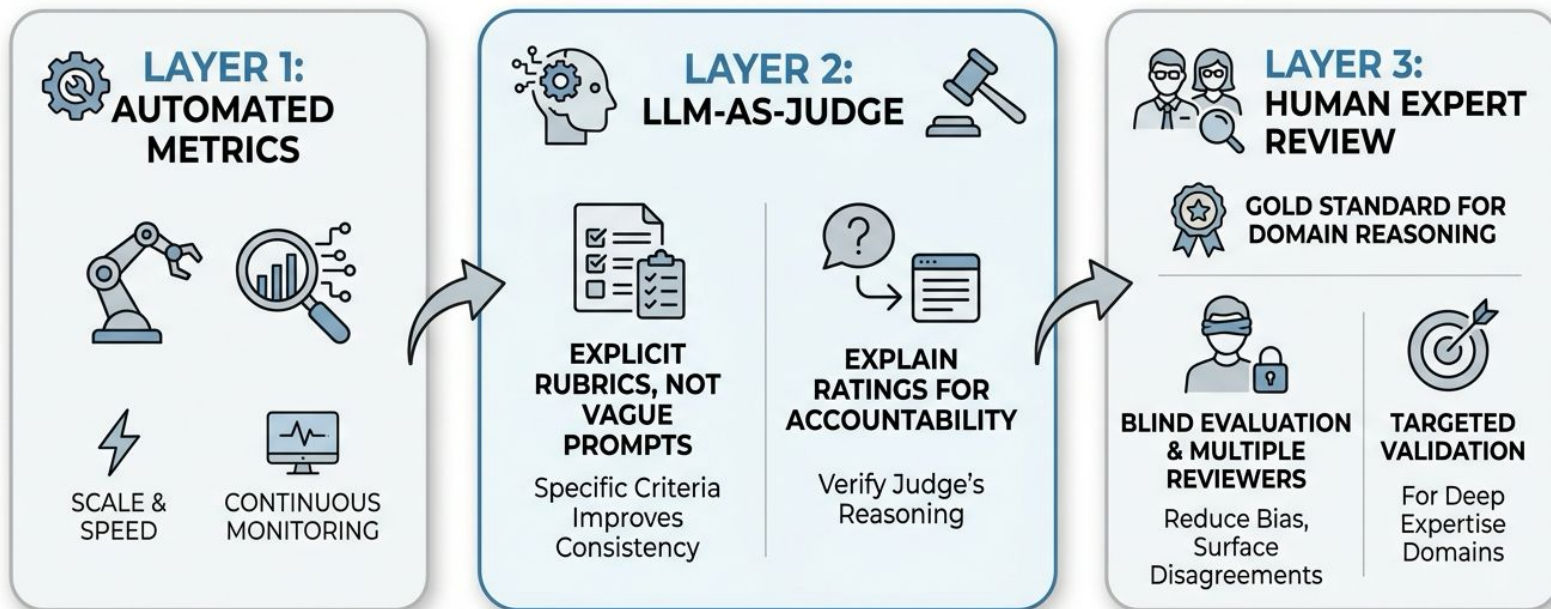
Self-Evaluation and Dual-Pass Architecture



Automated Metrics for Reasoning Quality

- Intra-step correctness rate: percent of logically valid steps
 - Inter-step consistency: 1 minus max contradiction probability
 - Informativeness score via conditional Predictive Value Index
 - Reasoning relevancy: percent of steps addressing user goals
 - Composite score uses minimum, not average, aggregation
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LLM-as-Judge and Human Evaluation



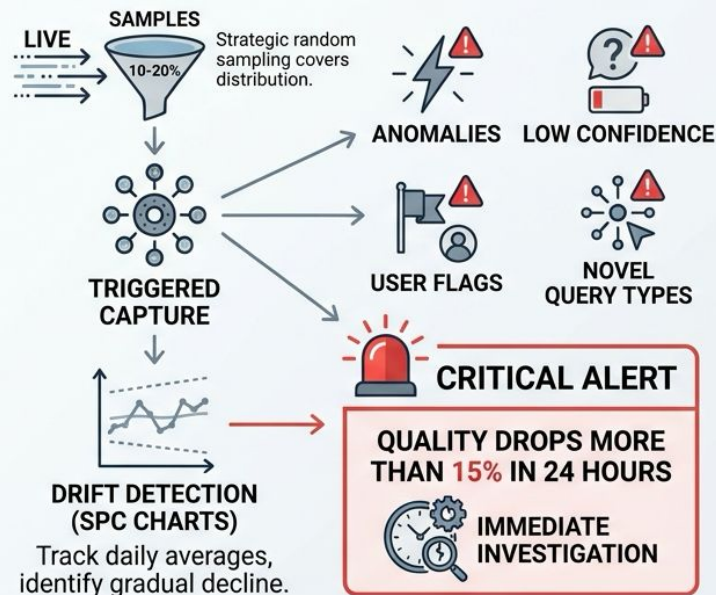
Production Architecture -- Offline and Online

OFFLINE PIPELINE



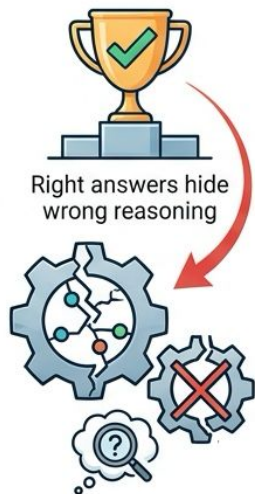
Structured reasoning traces, domain expert annotations, full metric suite assessment.

ONLINE MONITORING



Common Pitfalls

TASK SUCCESS ILLUSION



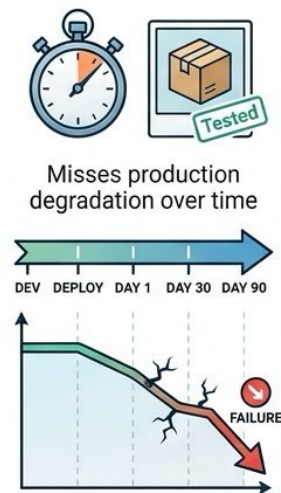
TRANSPARENCY-CONFLATION



MONOLITHIC METRICS



STATIC EVALUATION



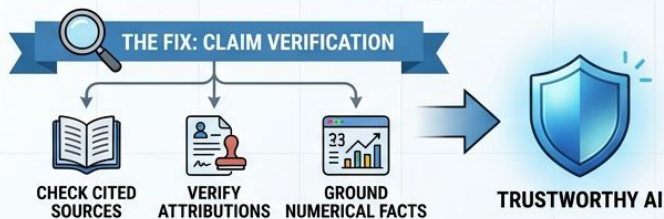
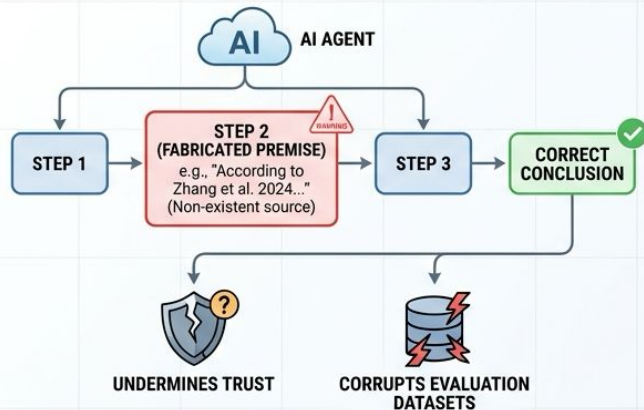
TESTING ONLY HAPPY PATHS

Ignores edge-case reasoning collapse

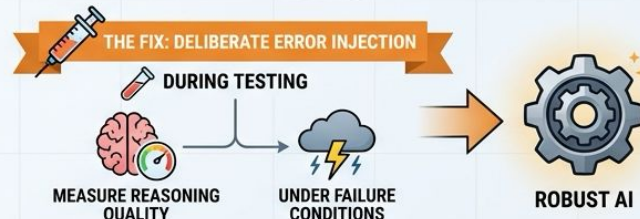
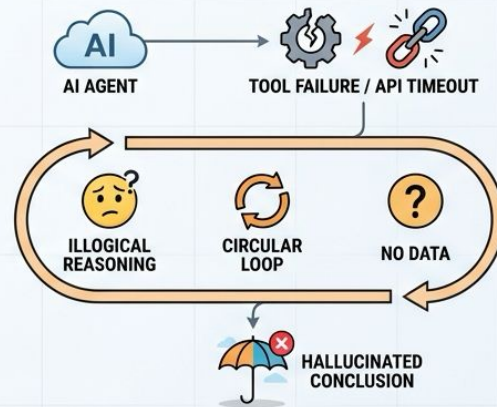


Blindspots

BLINDSPOT 1: HALLUCINATED REASONING TRACES



BLINDSPOT 2: ERROR-CONDITION REASONING LOOPS



Key Takeaways

- Reasoning quality predicts robustness beyond accuracy
 - Transparency enables trust, verification, and calibrated oversight
 - Multi-agent systems need system-level coherence evaluation
 - Continuous monitoring catches degradation before users notice
 - Next: Chapter 3.10 measures computational efficiency
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